

Nonequilibrium scheme for computing the flux of the convection-diffusion equation in the framework of the lattice Boltzmann method

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In this paper, we propose a local nonequilibrium scheme for computing the flux of the convection-diffusion equation with a source term in the framework of the multiple-relaxation-time (MRT) lattice Boltzmann method (LBM). Both the Chapman-Enskog analysis and the numerical results show that, at the diffusive scaling, the present nonequilibrium scheme has a second-order convergence rate in space. A comparison between the nonequilibrium scheme and the conventional second-order central-difference scheme indicates that, although both schemes have a second-order convergence rate in space, the present nonequilibrium scheme is more accurate than the central-difference scheme. In addition, the flux computation rendered by the present scheme also preserves the parallel computation feature of the LBM, making the scheme more efficient than conventional finite-difference schemes in the study of large-scale problems. Finally, a comparison between the single-relaxation-time model and the MRT model is also conducted, and the results show that the MRT model is more accurate than the single-relaxation-time model, both in solving the convection-diffusion equation and in computing the flux.

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I. INTRODUCTION

The convection-diffusion equation (CDE), describing heat and mass transfer, is given by [1]

$$\partial_t C + \nabla \cdot \mathbf{J} = R, \quad (1)$$

where C is a scalar variable and usually a function of both time and space, R is the source term, and $\mathbf{J}$, representing the flux, is defined by

$$\mathbf{J} = -D\nabla C + C\mathbf{u}, \quad (2)$$

with D denoting the diffusion coefficient and $\mathbf{u}$ denoting the velocity. For incompressible flows, the velocity can be obtained from the Navier-Stokes (N-S) equations,

$$\nabla \cdot \mathbf{u} = 0, \quad (3)$$

$$\partial_t \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla P + \nu \nabla^2 \mathbf{u} + \mathbf{a}, \quad (4)$$

where $P = p/\rho_0$ with p and ρ_0 representing the pressure and the approximately constant fluid density, ν is the kinematic viscosity, and $\mathbf{a}$ is the acceleration due to an external force.

Over the past decades, the lattice Boltzmann equation (LBM), as a mesoscopic numerical method based on the simplified kinetic models, has gained great success in the study of various fluid flow systems with complex physical phenomena and geometry structures [2–7] and has also been applied to solve the CDE [8–23]. It is noticed that most of the previous investigations on the CDE with the LBM only focused on the accuracy and convergence rate of the LBM in depicting the scalar variable C [8–23]. Recently, the computation of the flux has also received increasing attention in the study of heat transfer in turbulent Rayleigh-Bénard convection [24,25] and in porous media with an emphasis on obtaining effective physical properties of fluids in a porous medium [26,27].

In this paper, in the framework of the multiple-relaxation-time (MRT) LBM, we propose a local computation scheme (nonequilibrium scheme) for the flux of the CDE with a source term and show that the present scheme is more accurate than the nonlocal second-order central-difference schemes.

In the following, we first present a MRT lattice Boltzmann model for the CDE with a source term and construct a nonequilibrium scheme for the flux. We then test the accuracies and convergence rates of the MRT model for the CDE and the nonequilibrium scheme in computing the flux through some benchmark problems and demonstrate that both the MRT model and the nonequilibrium scheme have a second-order convergence rate in space. Finally, a comparison between the nonequilibrium scheme and the second-order central-difference scheme shows that the proposed nonequilibrium scheme is more accurate than the central-difference scheme in computing the flux.

II. MULTIPLE-RELAXATION-TIME LATTICE BOLTZMANN MODEL FOR THE CONVECTION-DIFFUSION EQUATION WITH A SOURCE TERM

In general, the models of the LBM can be classified into three kinds based on the collision operator: the single-relaxation-time model [or the Bhatnagar-Gross-Krook (BGK) model] [8–16,28], the two-relaxation-time (TRT) model [17–19,29], and the MRT model (or generalized model) [20–23,30,31]. In this paper, the MRT model is considered for its superiority over the BGK model in terms of stability and accuracy [21,32]. Although some MRT models for the CDE have been proposed over the past decade [17,20–22], their applications are limited by some assumptions. Ginzburg [17] and Rasin *et al.* [20] first extended the MRT model to the CDE and found that a numerical diffusion coefficient was induced in the recovered CDE when a linear equilibrium distribution function was used. Ginzburg also reported that the numerical diffusion

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coefficient proportional to the square of the velocity could be eliminated by adopting a nonlinear equilibrium distribution function, but the assumption $R \propto \delta^2$ (δ is a small parameter in the Chapman-Enskog analysis) was used to derive the correct CDE [17]. Yoshida and Nagaoka [21] also developed a MRT model for an anisotropic convection-diffusion equation (ACDE) with a linear equilibrium distribution function and found that, through an asymptotic analysis, the ACDE can be recovered at the diffusive scaling. Recently, Li *et al.* [22] extended the MRT model [21] to the axisymmetric CDE. In the following, we present a MRT model based on our previous paper [16] and show that the CDE with a source term can be recovered correctly through Chapman-Enskog analysis with no diffusive scaling as required in previous studies [21,22].

The evolution equation of the MRT model for the CDE, given by Eq. (1), can be written as [7,30,31]

$$g_i(\mathbf{x} + \mathbf{c}_i \delta t, t + \delta t) - g_i(\mathbf{x}, t) = \Omega_i + \delta t G_i, \quad (5)$$

where the collision operator Ω_i is defined by

$$\Omega_i = -(\mathbf{M}^{-1} \mathbf{\Lambda} \mathbf{M})_{ij} [g_j(\mathbf{x}, t) - g_j^{(\text{eq})}(\mathbf{x}, t)]. \quad (6)$$

$g_i(\mathbf{x}, t)$ is the distribution function associated with the discrete velocity $\mathbf{c}_i$ at position $\mathbf{x}$ and time t , $g_i^{(\text{eq})}(\mathbf{x}, t)$ is the equilibrium distribution function and is defined as [16]

$$g_i^{(\text{eq})} = w_i C \left[1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right] + \eta_i \frac{C P}{c_s^2}, \quad (7)$$

where the weight coefficients w_i and η_i for the D2Q9 model can be given by

$$\begin{aligned} w_0 &= \frac{4}{9}, \quad \eta_0 = -\frac{5}{9}, \\ w_{1-4} &= \eta_{1-4} = \frac{1}{9}, \quad w_{5-8} = \eta_{5-8} = \frac{4}{9}, \end{aligned} \quad (8)$$

and the discrete velocities $\mathbf{c}_i$ are defined as

$$\mathbf{c}_i = \begin{cases} (0, 0), & i = 0, \\ \{\cos[(2i-9)\pi/2], \sin[(i-1)\pi/2]\}c, & i = 1-4, \\ \{\cos[(2i-9)\pi/4], \sin[(2i-9)\pi/4]\}\sqrt{2}c, & i = 5-8, \end{cases} \quad (9)$$

where $c = \delta x / \delta t = 1$ with δx and δt representing the lattice spacing and time step and $c_s = 1/\sqrt{3}c$. The transformation matrix $\mathbf{M}$ in Eq. (6) is given by

$$\mathbf{M} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -4 & -1 & -1 & -1 & 2 & 2 & 2 & 2 & 2 \\ 4 & -2 & -2 & -2 & -2 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & -2 & 0 & 2 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \\ 0 & 0 & -2 & 0 & 2 & 1 & 1 & -1 & -1 \\ 0 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \end{pmatrix}, \quad (10)$$

which is constructed based on the polynomials of the discrete velocity $\mathbf{c}_i$ and can be used to project the distribution function

$g_i(\mathbf{x}, t)$ and $g_i^{(\text{eq})}(\mathbf{x}, t)$ in the discrete velocity space onto macroscopic variables in the moment space [31],

$$\mathbf{m} := \mathbf{M} \mathbf{g}, \quad \mathbf{m}^{(\text{eq})} := \mathbf{M} \mathbf{g}^{(\text{eq})}, \quad (11)$$

where $\mathbf{m} = (m_0, m_1, \dots, m_8)^T$, $\mathbf{m}^{(\text{eq})} = (m_0^{(\text{eq})}, m_1^{(\text{eq})}, \dots, m_8^{(\text{eq})})^T$, $\mathbf{g} = (g_0, g_1, \dots, g_8)^T$, and $\mathbf{g}^{(\text{eq})} = (g_0^{(\text{eq})}, g_1^{(\text{eq})}, \dots, g_8^{(\text{eq})})^T$. $\mathbf{\Lambda}$ is a non-negative and diagonal relaxation matrix and can be written in the following form:

$$\mathbf{\Lambda} = \text{diag}(\lambda_0, \lambda_1, \dots, \lambda_8), \quad (12)$$

where $0 < \lambda_i < 2$.

To derive the correct CDE from the present MRT model, the discrete source term G_i is defined as

$$G_i = \left[\mathbf{M}^{-1} \left(\mathbf{I} - \frac{\mathbf{\Lambda}}{2} \right) \mathbf{M} \right]_{ij} R_j, \quad (13)$$

where R_i is related to the source term R based on

$$R_i = w_i \left[\left(1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} \right) R + \frac{\mathbf{c}_i \cdot (P \nabla C + C \mathbf{a})}{c_s^2} \right]. \quad (14)$$

The macroscopic scalar variable C should be simultaneously computed by

$$C = \sum_i g_i + \frac{\delta t}{2} R. \quad (15)$$

It is worth noting that the present discrete source term, given by Eq. (13), and the computational scheme for scalar variable C , given by Eq. (15), are similar to those used for the flow field [7,33–36].

We now present the Chapman-Enskog analysis to derive the CDE from the MRT model. We first expand the distribution function, the time and space derivatives, the source term R , and the acceleration $\mathbf{a}$ as

$$g_i = g_i^{(0)} + \delta g_i^{(1)} + \delta^2 g_i^{(2)} + \dots, \quad (16a)$$

$$\partial_t = \delta \partial_{t_1} + \delta^2 \partial_{t_2}, \quad (16b)$$

$$\partial_x = \delta \partial_{x_1}, \quad (16c)$$

$$R = \delta R_1, \quad \mathbf{a} = \delta \mathbf{a}_1, \quad (16d)$$

where δ is a small parameter and $\mathbf{a}_1 = (a_{x_1}, a_{y_1})$.

Substituting Eqs. (16) into Eq. (5) and adopting the Taylor expansion, we obtain the zero-, first-, and second-order equations in δ ,

$$\delta^0: g_i^{(0)} = g_i^{(\text{eq})}, \quad (17a)$$

$$\begin{aligned} \delta^1: & (\partial_{t_1} + \mathbf{c}_i \cdot \nabla_1) g_i^{(0)} \\ &= -\frac{1}{\delta t} (\mathbf{M}^{-1} \mathbf{\Lambda} \mathbf{M})_{ij} g_j^{(1)} + \left[\mathbf{M}^{-1} \left(\mathbf{I} - \frac{\mathbf{\Lambda}}{2} \right) \mathbf{M} \right]_{ij} R_{1,j}, \end{aligned} \quad (17b)$$

$$\begin{aligned} \delta^2: & \partial_{t_2} g_i^{(0)} + (\partial_{t_1} + \mathbf{c}_i \cdot \nabla_1) g_i^{(1)} + \frac{\delta t}{2} (\partial_{t_1} + \mathbf{c}_i \cdot \nabla_1)^2 g_i^{(0)} \\ &= -\frac{1}{\delta t} (\mathbf{M}^{-1} \mathbf{\Lambda} \mathbf{M})_{ij} g_j^{(2)}, \end{aligned} \quad (17c)$$

where $R_{1,i} = w_i \left[\left(1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} \right) R_1 + \frac{\mathbf{c}_i \cdot (P \nabla_1 C + C \mathbf{a}_1)}{c_s^2} \right]$.

Rewriting Eqs. (17) in vector form and multiplying $\mathbf{M}$ on both sides of them, one can easily obtain the equations in the moment space,

$$\delta^0: \mathbf{m}^{(0)} = \mathbf{m}^{(\text{eq})}, \quad (18a)$$

$$\delta^1: \mathbf{D}_1 \mathbf{m}^{(0)} = -\frac{1}{\delta t} \mathbf{\Lambda} \mathbf{m}^{(1)} + \left(\mathbf{I} - \frac{\mathbf{\Lambda}}{2} \right) \mathbf{M} \mathbf{R}_1, \quad (18b)$$

$$\delta^2: \frac{\partial \mathbf{m}^{(0)}}{\partial t_2} + \mathbf{D}_1 \left(\mathbf{I} - \frac{\mathbf{\Lambda}}{2} \right) \mathbf{m}^{(1)} + \frac{\delta t}{2} \mathbf{D}_1 \left(\mathbf{I} - \frac{\mathbf{\Lambda}}{2} \right) \mathbf{M} \mathbf{R}_1 = -\frac{1}{\delta t} \mathbf{\Lambda} \mathbf{m}^{(2)}, \quad (18c)$$

where $\mathbf{D}_1 = \partial_{t_1} \mathbf{I} + \mathbf{M} \partial_{\alpha_1} \text{diag}(c_{0,\alpha}, \dots, c_{8,\alpha}) \mathbf{M}^{-1}$, $\mathbf{m}^{(1)} = \mathbf{M} \mathbf{g}^{(1)} / \delta = (m_0^{(1)}, \dots, m_8^{(1)})^T$, $\mathbf{g}^{(1)} = (g_0^{(1)}, \dots, g_8^{(1)})^T$, and $\mathbf{R}_1 = (R_{1,0}, \dots, R_{1,8})^T$. According to Eq. (18b), we can rewrite the first-order equations in δ , but we only present the first, fourth, and sixth ones since only these three equations

are useful in deriving the CDE,

$$\partial_{t_1} C + \partial_{x_1} (C u_x) + \partial_{y_1} (C u_y) = R_1, \quad (19a)$$

$$\begin{aligned} \partial_{t_1} C u_x + \partial_{x_1} [C(c_s^2 + P + u_x^2)] + \partial_{y_1} (C u_x u_y) \\ = -\frac{\lambda_3}{\delta t} m_3^{(1)} + \left(1 - \frac{\lambda_3}{2} \right) (P \partial_{x_1} C + C a_{x_1} + u_x R_1), \end{aligned} \quad (19b)$$

$$\begin{aligned} \partial_{t_1} C u_y + \partial_{x_1} (C u_x u_y) + \partial_{y_1} [C(c_s^2 + P + u_y^2)] \\ = -\frac{\lambda_5}{\delta t} m_5^{(1)} + \left(1 - \frac{\lambda_5}{2} \right) (P \partial_{y_1} C + C a_{y_1} + u_y R_1), \end{aligned} \quad (19c)$$

where Eq. (15) has been used to derive Eq. (19a). Similarly, based on Eq. (18c) we can also give the second-order equations in δ , but only the one corresponding to the conservative variable C is given

$$\partial_{t_2} C + \partial_{x_1} \left\{ \left(1 - \frac{\lambda_3}{2} \right) \left[m_3^{(1)} + \frac{\delta t}{2} (P \partial_{x_1} C + C a_{x_1} + u_x R_1) \right] \right\} + \partial_{y_1} \left\{ \left(1 - \frac{\lambda_5}{2} \right) \left[m_5^{(1)} + \frac{\delta t}{2} (P \partial_{y_1} C + C a_{y_1} + u_y R_1) \right] \right\} = 0. \quad (20)$$

From Eqs. (19b) and (19c), we obtain

$$\begin{aligned} m_3^{(1)} + \frac{\delta t}{2} (P \partial_{x_1} C + C a_{x_1} + u_x R_1) &= -\frac{\delta t}{\lambda_3} [\partial_{t_1} (C u_x) + \partial_{x_1} (C c_s^2 + C u_x^2 + C P) + \partial_{y_1} (C u_x u_y) - (P \partial_{x_1} C + C a_{x_1} + u_x R_1)] \\ &= -\frac{\delta t}{\lambda_3} [C(\partial_{t_1} u_x + u_x \partial_{x_1} u_x + u_y \partial_{y_1} u_x + \partial_{x_1} P - a_{x_1}) + \partial_{x_1} (C c_s^2)], \end{aligned} \quad (21)$$

$$\begin{aligned} m_5^{(1)} + \frac{\delta t}{2} (P \partial_{y_1} C + C a_{y_1} + u_y R_1) &= -\frac{\delta t}{\lambda_5} [\partial_{t_1} (C u_y) + \partial_{x_1} (C u_x u_y) + \partial_{y_1} (C c_s^2 + C u_y^2 + C P) - (P \partial_{y_1} C + C a_{y_1} + u_y R_1)] \\ &= -\frac{\delta t}{\lambda_5} [C(\partial_{t_1} u_y + u_x \partial_{x_1} u_y + u_y \partial_{y_1} u_y + \partial_{y_1} P - a_{y_1}) + \partial_{y_1} (C c_s^2)]. \end{aligned} \quad (22)$$

With the help of the first-order incompressible Navier-Stokes equations in δ [37,38],

$$\partial_{t_1} \mathbf{u} + \mathbf{u} \cdot \nabla_1 \mathbf{u} = -\nabla_1 P + \mathbf{a}_1, \quad (23)$$

one can simplify Eqs. (21) and (22) to

$$m_3^{(1)} + \frac{\delta t}{2} (P \partial_{x_1} C + C a_{x_1} + u_x R_1) = -\frac{\delta t}{\lambda_3} \partial_{x_1} (C c_s^2), \quad (24)$$

$$m_5^{(1)} + \frac{\delta t}{2} (P \partial_{y_1} C + C a_{y_1} + u_y R_1) = -\frac{\delta t}{\lambda_5} \partial_{y_1} (C c_s^2). \quad (25)$$

Substituting Eqs. (24) and (25) into Eq. (20), we have

$$\begin{aligned} \partial_{t_2} C + \partial_{x_1} \left[\left(\frac{1}{2} - \frac{1}{\lambda_3} \right) c_s^2 \delta t \partial_{x_1} C \right] \\ + \partial_{y_1} \left[\left(\frac{1}{2} - \frac{1}{\lambda_5} \right) c_s^2 \delta t \partial_{y_1} C \right] = 0. \end{aligned} \quad (26)$$

Combining Eq. (19a) at the t_1 time scale and Eq. (26) at the t_2 time scale, we arrive at the CDE as

$$\partial_t C + \nabla \cdot (C \mathbf{u}) = D \nabla^2 C + R, \quad (27)$$

where the relation $\lambda_D := \lambda_3 = \lambda_5$ is used to derive the isotropic diffusion coefficient D ,

$$D = c_s^2 \left(\frac{1}{\lambda_D} - \frac{1}{2} \right) \delta t. \quad (28)$$

From the above procedure, it is clear that the CDE with a source term can be recovered correctly from the present MRT model without adopting any approximations. In addition, although a space derivative appears in the discrete source term, given by Eq. (14), it can be computed locally in the LBM. The computation details can be found elsewhere [16]. Therefore, the collision process of the present model can be conducted locally. In addition to this feature, such as the two-relaxation-time model [17], the present MRT model can also be used to study the anisotropic CDE when $\lambda_3 \neq \lambda_5$. Other features of the present MRT model are discussed as follows:

Remark I. If the parameters λ_i of the relaxation matrix are equal to each other and are denoted by λ_{BGK} , the MRT model reduces to the BGK model with a simpler form

$$\begin{aligned} g_i(\mathbf{x} + \mathbf{c}_i \delta t, t + \delta t) - g_i(\mathbf{x}, t) \\ = -\lambda_{\text{BGK}}[g_i(\mathbf{x}, t) - g_i^{(\text{eq})}(\mathbf{x}, t)] + \delta t \left(1 - \frac{\lambda_{\text{BGK}}}{2}\right) R_i, \end{aligned} \quad (29)$$

where R_i is still given by Eq. (14).

Remark II. Historically, there are several discrete source term schemes that have been adopted in the LBM for the CDE with a source term [8,11,12,17,21,39,40]. Dawson *et al.* [8] first proposed a simple discrete source term scheme with the form $R_i = w_i R$, but the assumption $R \propto \delta^2$ was used to derive the correct CDE. We note that this assumption is also adopted in the subsequently developed TRT and MRT models [17]. However, this assumption may be not satisfied and may lead to large errors in practice as discussed elsewhere [11]. In order to eliminate the assumption $R \propto \delta^2$, Deng *et al.* [11] and Shi *et al.* [12] proposed a more general discrete source scheme related to time and/or the space derivative of source term R to ensure that the CDE can be recovered under the assumption of constant velocity, but additional finite-difference schemes may be needed to compute the time and space derivatives. Recently, to avoid the use of the time or space derivative, Seta [39] used another scheme given by Wang *et al.* [40],

$$G_i = w_i \left(1 - \frac{\lambda_{\text{BGK}}}{2}\right) R, \quad (30)$$

and redefined a new computational formula, the same as Eq. (15), to compute scalar variable C [39,40]. However, as shown in the paper, to derive the correct CDE, the terms $O(Cu^2)$ are neglected in the Chapman-Enskog analysis, whereas they are retained in the equilibrium distribution function. In addition to this contradiction, we can also find that, even for the case with a constant velocity, the term $O(Ru)$ must be neglected if one wants to derive the correct CDE [39]. In this paper, a discrete source term scheme [Eq. (13)] and a formula for scalar variable C [Eq. (15)] are proposed to ensure that the CDE can be recovered correctly from the present model, and simultaneously, the problems mentioned above can be solved completely.

Remark III. Based on the above Chapman-Enskog analysis, one can also find that, for the case with a constant velocity, as considered by Deng *et al.* [11] and Shi *et al.* [12], the present MRT model can be further simplified with the following equilibrium distribution function $g_i^{(\text{eq})}$ and R_i :

$$g_i^{(\text{eq})} = w_i C \left[1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2}\right], \quad (31)$$

$$R_i = w_i R \left(1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2}\right), \quad (32)$$

where, for simplicity, pressure P and acceleration $\mathbf{a}$ are set to be zero. In addition, we also note that, for this special case, the

discrete source term scheme for the BGK model should read

$$G_i = w_i R \left(1 - \frac{\lambda_{\text{BGK}}}{2}\right) \left(1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2}\right), \quad (33)$$

which can also be derived from Ref. [12] once the following transformation is adopted for the case of $\theta = 1$ (see Eq. (19) in Ref. [12]),

$$g_i(\mathbf{x}, t) = g_i^s(\mathbf{x}, t) - \frac{\delta t}{2} R_i, \quad (34)$$

which can be used to eliminate the implicitness of the LBM [34,40,41], and $g_i^s(\mathbf{x}, t)$ is the distribution function used in the previous paper [12].

III. THE FLUX COMPUTATION IN THE MULTIPLE-RELAXATION-TIME LATTICE BOLTZMANN MODEL

In this section, we present a local computational scheme for the flux in the framework of the LBM. Based on Eqs. (24) and (25), we can obtain

$$m_3^{(1)} + \frac{\delta t}{2} (Ca_{x_1} + u_x R_1) = -\delta t \left(\frac{c_s^2}{\lambda_D} + \frac{P}{2}\right) \partial_{x_1} C, \quad (35)$$

$$m_5^{(1)} + \frac{\delta t}{2} (Ca_{y_1} + u_y R_1) = -\delta t \left(\frac{c_s^2}{\lambda_D} + \frac{P}{2}\right) \partial_{y_1} C, \quad (36)$$

where the relation $\lambda_D = \lambda_3 = \lambda_5$ is used for the isotropic problem considered here. From Eqs. (35) and (36), one can further obtain

$$\partial_{x_1} C = -\frac{m_3^{(1)} + \frac{\delta t}{2} (Ca_{x_1} + u_x R_1)}{\delta t \left(\frac{c_s^2}{\lambda_D} + \frac{P}{2}\right)}, \quad (37)$$

$$\partial_{y_1} C = -\frac{m_5^{(1)} + \frac{\delta t}{2} (Ca_{y_1} + u_y R_1)}{\delta t \left(\frac{c_s^2}{\lambda_D} + \frac{P}{2}\right)}. \quad (38)$$

According to Eqs. (10), (11), and (16a), the nonequilibrium terms $m_3^{(1)}$ and $m_5^{(1)}$ can be given as

$$m_3^{(1)} = \sum_i c_{i,x} g_i^{(1)}, \quad m_5^{(1)} = \sum_i c_{i,y} g_i^{(1)}. \quad (39)$$

Multiplying δ on the both sides of Eqs. (35) and (36) and using the approximation $\delta g_i^{(1)} \approx g_i - g_i^{(\text{eq})}$, we can obtain a computational scheme for the gradient term $\nabla C = (\partial_x C, \partial_y C)$,

$$\nabla C = -\frac{\sum_i \mathbf{c}_i [g_i - g_i^{(\text{eq})}] + \frac{\delta t}{2} (C\mathbf{a} + \mathbf{u}R)}{\delta t \left(\frac{c_s^2}{\lambda_D} + \frac{P}{2}\right)}. \quad (40)$$

It can be seen that Eq. (40) consists of a local computational scheme for the gradient term ∇C since the right hand side terms can be computed directly in the LBM without adopting any finite-difference schemes. Based on the results, the term R_i [Eq. (14)] can also be calculated locally, and consequently, the collision process of the present MRT model can be performed locally, which is important to preserve some advantages of the LBM in the study of the heat and/or mass transfer in complex geometries [16].

Substituting Eq. (40) into Eq. (2), we can obtain a local computational scheme for the flux,

$$\mathbf{J} = D \frac{\sum_i \mathbf{c}_i [g_i - g_i^{(\text{eq})}] + \frac{\delta t}{2} (C\mathbf{a} + \mathbf{u}R)}{\delta t \left(\frac{c_s^2}{\lambda_D} + \frac{P}{2} \right)} + C\mathbf{u}, \quad (41)$$

or equivalently,

$$J_\alpha = D \frac{(\mathbf{m} - \mathbf{m}^{(\text{eq})})_\beta + \frac{\delta t}{2} (C a_\alpha + \mathbf{u}_\alpha R)}{\delta t \left(\frac{c_s^2}{\lambda_D} + \frac{P}{2} \right)} + C u_\alpha, \quad (42)$$

where $\alpha = x, \beta = 3$ or $\alpha = y, \beta = 5$. We note that the present local computational scheme for flux is similar to the previous result reported elsewhere [16] where the source term effect is not included. From Eqs. (41) and (42) we can also find that, in the same way as the computation of the shear stress [36,42–44], the flux can be computed locally by the nonequilibrium part of the distribution function, and for this reason, Eq. (41) or Eq. (42) is also called the nonequilibrium scheme for the flux. Some remarks on the local nonequilibrium scheme are as follows:

Remark IV. Based on Eq. (41) or Eq. (42), it is interesting to note that, for the isotropic CDE considered here, the computation of flux in the MRT model is the same as that in the BGK model. In addition, one can also find that both acceleration $\mathbf{a}$ due to the external force and source term R have an influence on the computation of flux in the LBM.

Remark V. For the special case with a constant velocity, the nonequilibrium scheme for the flux can be simplified through setting pressure P and acceleration $\mathbf{a}$ to be zero,

$$\mathbf{J} = \left(1 - \frac{\lambda_D}{2}\right) \left[\sum_i \mathbf{c}_i (g_i - g_i^{(\text{eq})}) + \frac{\delta t}{2} \mathbf{u}R \right] + C\mathbf{u}. \quad (43)$$

Remark VI. For a pure isotropic diffusion process, all physical variables related to the flow field, including velocity $\mathbf{u}$, pressure P , and acceleration $\mathbf{a}$, can be set to be zero, and consequently, the nonequilibrium scheme for the flux [Eq. (42)] can be further simplified

$$\mathbf{J} = \left(1 - \frac{\lambda_D}{2}\right) \sum_i \mathbf{c}_i [g_i - g_i^{(\text{eq})}], \quad (44)$$

which has been reported previously [16,26,27].

In the following, we discuss the convergence rate of the nonequilibrium method in computing the flux. As discussed elsewhere [7,16,18,45], both lattice Boltzmann models for scalar variable C and velocity $\mathbf{u}$ have a second-order convergence rate in space, as a result, the convective part of the flux, $C\mathbf{u}$ in Eq. (42), should have a second-order accuracy. We now give an evaluation on the first term on the right hand side of Eq. (41), i.e., the diffusive part of the flux,

$$\mathbf{J}_d = D \frac{\sum_i \mathbf{c}_i [g_i - g_i^{(\text{eq})}] + \frac{\delta t}{2} (C\mathbf{a} + \mathbf{u}R)}{\delta t \left(\frac{c_s^2}{\lambda_D} + \frac{P}{2} \right)}. \quad (45)$$

Based on Eqs. (37) and (38), one can find that the diffusive flux can be computed with a second-order convergence rate

once $g_i^{(1)}$ is known, but unfortunately, $g_i^{(1)}$ cannot be calculated directly in the LBM, which is also the reason why we use $g_i^{(\text{neq})}$ ($= g_i - g_i^{(\text{eq})}$) instead of $g_i^{(1)}$ to compute the diffusive flux, as shown above. However, it should be noted that the use of the approximation $\delta g_i^{(1)} \approx g_i^{(\text{neq})}$ in the computation of the diffusive flux may cause some errors. To check the errors caused by the approximation, we need to evaluate the influence of higher-order terms of the nonequilibrium part ($g_i^{(k)}, k \geq 2$) on the diffusive flux. But in the following analysis, only the second-order term of the nonequilibrium part $g_i^{(2)}$ is considered since the higher-order terms of the nonequilibrium part become less important with increasing order of δ [36,42]. At the diffusive scaling [21,42,43], the lattice spacing and time step satisfy the relation $\delta t \propto \delta x^2$, and consequently, we can derive the following relations:

$$\delta t c_s^2 \sim O(1), \quad \delta t \left(\frac{c_s^2}{\lambda_D} + \frac{P}{2} \right) \sim O(1), \quad (46)$$

$$\mathbf{J}_d \propto D \left\{ \sum_i \mathbf{c}_i [g_i - g_i^{(\text{eq})}] + \frac{\delta t}{2} (C\mathbf{a} + \mathbf{u}R) \right\}. \quad (47)$$

According to Eq. (18c), we can obtain

$$\begin{aligned} m_3^{(2)} &= \sum_i c_{i,x} g_i^{(2)} \\ &= -\frac{\delta t}{\lambda_3} \left\{ \partial_x m_5^{(0)} + \left[\mathbf{D}_1 \left(\mathbf{I} - \frac{\Lambda}{2} \right) \left(\mathbf{m}^{(1)} + \frac{\delta t}{2} \mathbf{M} \mathbf{R}_1 \right) \right]_3 \right\}, \end{aligned} \quad (48)$$

$$\begin{aligned} m_5^{(2)} &= \sum_i c_{i,y} g_i^{(2)} \\ &= -\frac{\delta t}{\lambda_5} \left\{ \partial_y m_5^{(0)} + \left[\mathbf{D}_1 \left(\mathbf{I} - \frac{\Lambda}{2} \right) \left(\mathbf{m}^{(1)} + \frac{\delta t}{2} \mathbf{M} \mathbf{R}_1 \right) \right]_5 \right\}. \end{aligned} \quad (49)$$

After some algebraic manipulations, one can readily show

$$m_3^{(2)} \propto \delta t \text{ or } \delta x^2, \quad m_5^{(2)} \propto \delta t \text{ or } \delta x^2. \quad (50)$$

Based on the above results and Eq. (45), we can further derive

$$\mathbf{J}_d = D \frac{\sum_i \mathbf{c}_i [g_i - g_i^{(\text{eq})}] + \frac{\delta t}{2} (C\mathbf{a} + \mathbf{u}R)}{\delta t \left(\frac{c_s^2}{\lambda_D} + \frac{P}{2} \right)} + O(\delta x^2), \quad (51)$$

which can also be expressed by the macroscopic variables in the moment space, similar to Eq. (42). From above analysis it can also be concluded that, at the diffusive scaling, the nonequilibrium scheme for the flux has a second-order convergence rate in space.

IV. NUMERICAL RESULTS AND DISCUSSION

This section discusses the accuracies and convergence rates of the present MRT model for the CDE and the nonequilibrium scheme for flux. The discussion is made by comparing the analytical solutions to three physical problems and numerical results obtained with the present model and the nonequilibrium scheme. The problems under consideration

include diffusion of the Gaussian hill [13,19,21], thermal Couette flow with heat dissipation [46–50], and planar thermal Poiseuille flow [50,51]. The relative error (E) is used to test the deviation between the analytical solutions and the numerical results and is defined as

$$E(\phi) = \frac{\sum_{\mathbf{x}} |\phi_a(\mathbf{x}, t) - \phi_n(\mathbf{x}, t)|}{\sum_{\mathbf{x}} |\phi_a(\mathbf{x}, t)|}, \quad (52)$$

where ϕ represents scalar variable C or one element of flux $\mathbf{J}$ and subscripts a and n denote the analytical and numerical solutions. In our simulations, the distribution function is initialized by its equilibrium distribution function, i.e.,

$$g_i(x, y, t = 0) = g_i^{(eq)}(x, y, t = 0). \quad (53)$$

Unless otherwise stated, the anti-bounce-back scheme [17,45] and halfway bounce-back scheme [4,7,46] are adopted for the boundary conditions of the CDE and the N-S equations due to fact that they are local approaches and have a second-order accuracy in space. In addition, besides the relaxation factors λ_i ($i = 3$ and 5) related to diffusion coefficient D , the other

free relaxation factors λ_i in this paper, for simplicity, are set as $\lambda_0 = 0, \lambda_1 = \lambda_2 = \lambda_4 = \lambda_6 = \lambda_7 = \lambda_8 = 1.0$.

A. Validations

1. Example 1. Diffusion of the Gaussian hill

Diffusion of the Gaussian hill is a simple but quite classic convection-diffusion problem and has been widely used to validate lattice Boltzmann models for the CDE without a source term [13,19,21]. For a given constant velocity $\mathbf{u}_0 = (U_0, V_0)$ and the following initial condition:

$$C(x, y, t = 0) = \frac{1}{10\pi} \exp\left(-\frac{x^2 + y^2}{10}\right), \quad (54)$$

the analytical solution of this problem is [19]

$$C(x, y, t) = \frac{1}{\pi(10 + 4Dt)} \exp\left(-\frac{(x - U_0 t)^2 + (y - V_0 t)^2}{10 + 4Dt}\right). \quad (55)$$

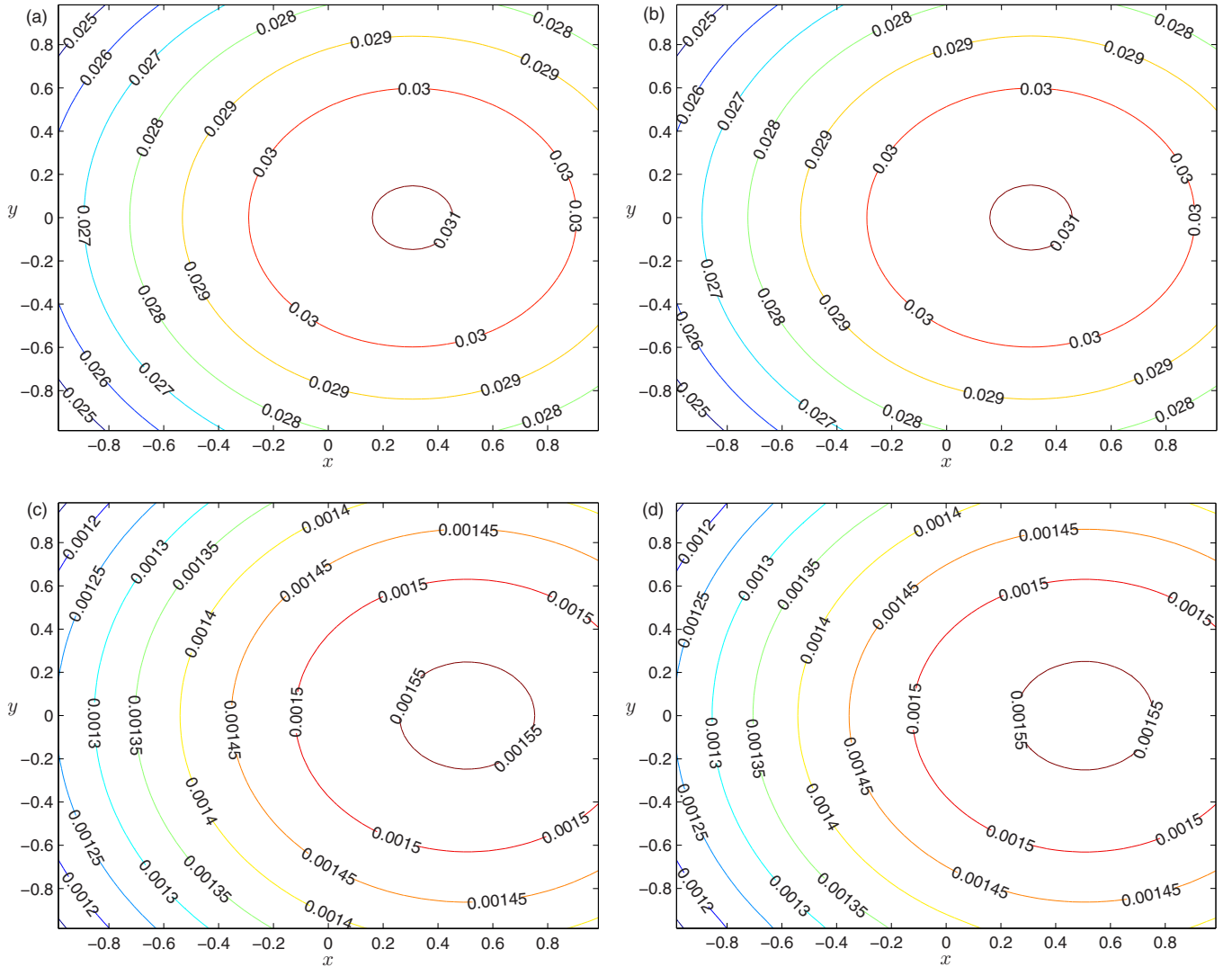


FIG. 1. (Color online) The contour lines of concentration C [(a) analytical solution and (b) numerical result] and flux J_x [(c) analytical solution and (d) numerical result] ($Pe = 10$).

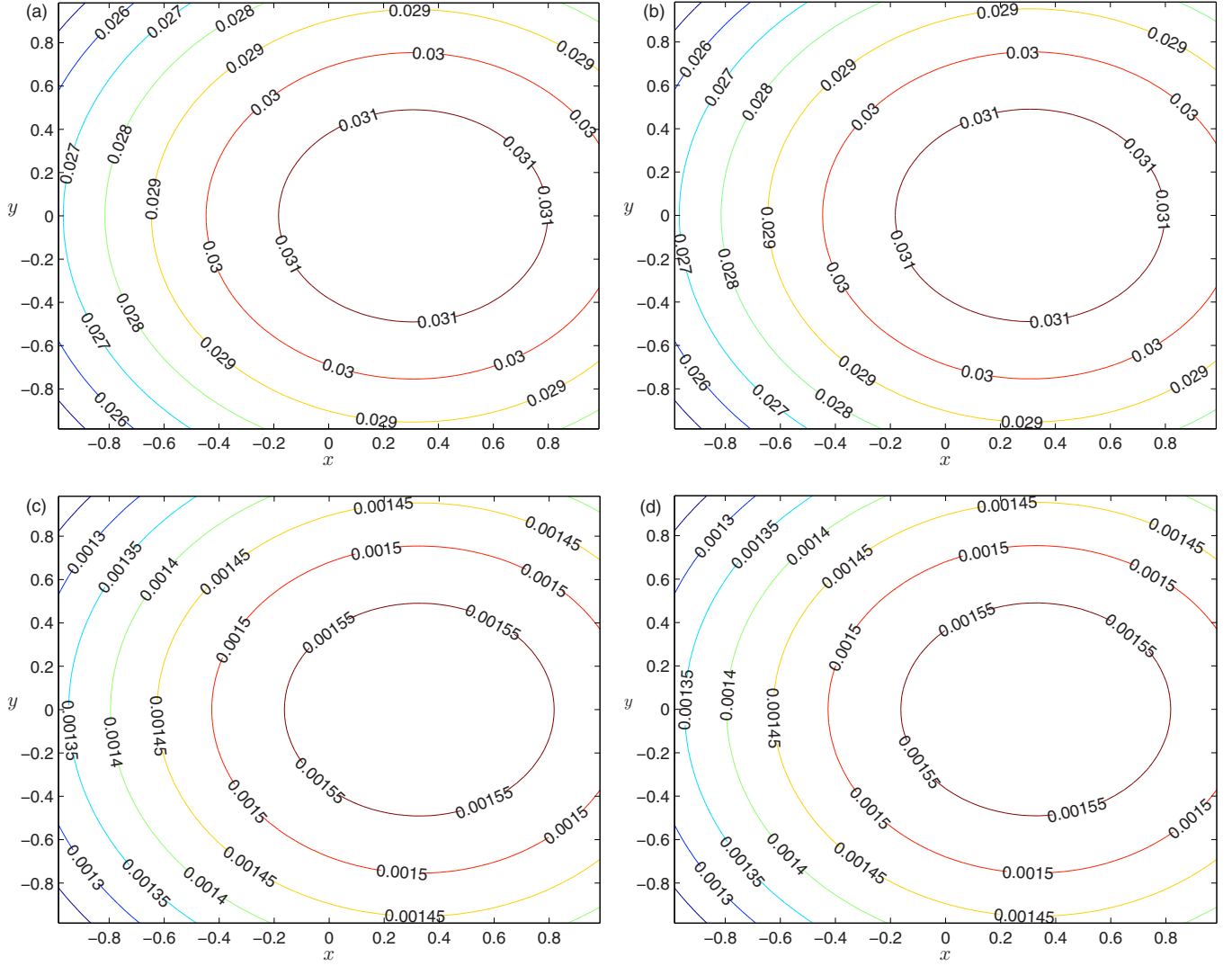


FIG. 2. (Color online) The contour lines of concentration C [(a) analytical results and (b) numerical results] and flux J_x [(c) analytical results and (d) numerical results] ($Pe = 100$).

Based on Eqs. (2) and (55), we can also derive flux $\mathbf{J} = (J_x, J_y)$,

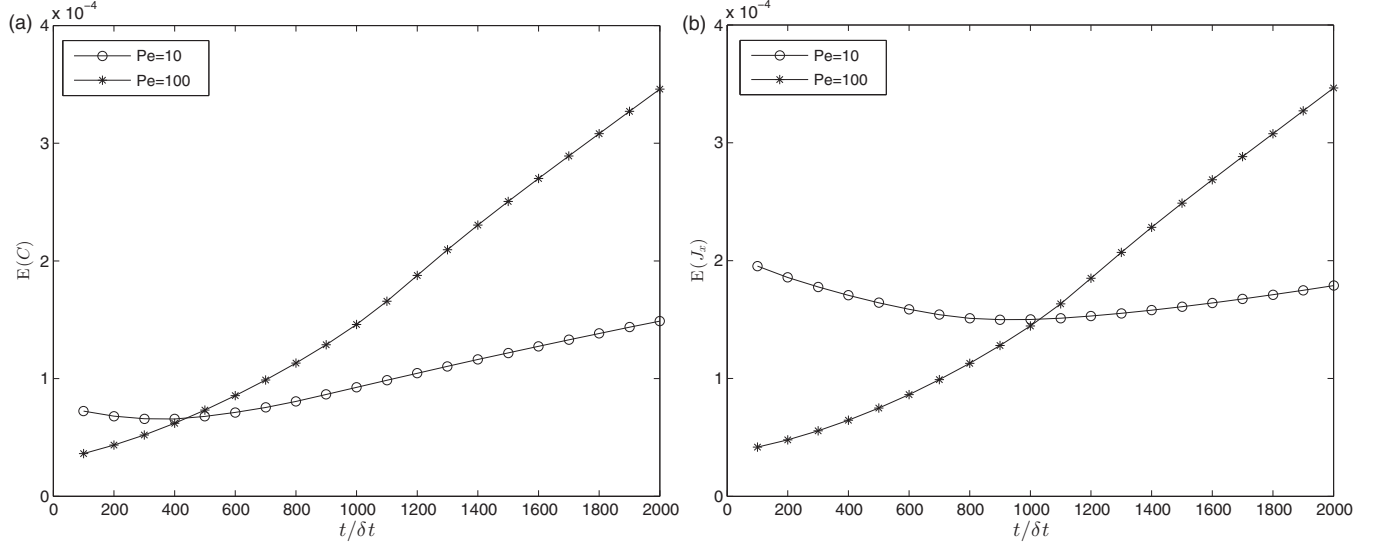
$$J_x = -D \frac{\partial C}{\partial x} + C U_0 = \frac{1}{\pi(10 + 4Dt)} \times \exp\left(-\frac{(x - U_0 t)^2 + (y - V_0 t)^2}{10 + 4Dt}\right) \times \left[\frac{D(x - U_0 t)}{\pi(5 + 2Dt)} + U_0\right], \quad (56)$$

$$J_y = -D \frac{\partial C}{\partial y} + C V_0 = \frac{1}{\pi(10 + 4Dt)} \times \exp\left(-\frac{(x - U_0 t)^2 + (y - V_0 t)^2}{10 + 4Dt}\right) \times \left[\frac{D(y - V_0 t)}{\pi(5 + 2Dt)} + V_0\right]. \quad (57)$$

The computational domain $x, y \in [-1, 1] \times [-1, 1]$ is considered, and the elements of velocities U_0 and V_0 in the x and y directions are set to be 0.05 and 0. The Dirichlet boundary conditions given based on Eq. (55) are adopted on boundaries in the x and y directions.

The numerical results for concentration C and flux J_x are presented in Figs. 1 and 2 where the Péclet number (Pe) is defined by $Pe = LU_0/D$, λ_D is determined according to Eq. (28), the grid number (N) is set to be 64, which is fine enough to give accurate results, and the simulations are suspended at $T = 200 \times \delta t$. As shown in these figures, the numerical results are very close to the corresponding analytical solutions.

In addition, we also note that as this problem is time dependent, the relative error may change with time. Thus, it is necessary to test the variation in relative error E [Eq. (52)] with time. To this end, we computed the relative errors of concentration C and flux J_x at different times and showed the results in Fig. 3. As seen from this figure, although the relative

FIG. 3. The relative errors of (a) concentration C and (b) flux J_x at different times.

errors vary with time, especially for the case of $Pe = 100$, generally their increments in time seem not very significant.

2. Example 2. Thermal Couette flow with viscous heat dissipation

Thermal Couette flow with viscous heat dissipation has also been used to test thermal lattice Boltzmann models and thermal boundary conditions [47–50]. The schematic of the problem is shown in Fig. 4 where the top wall moves with a constant velocity in a horizontal direction and at a higher temperature T_0 , whereas the bottom wall remains stationary and at a lower temperature T_1 . In order to keep the expressions and description on this thermal problem more consistent, hereafter the variable C in Eq. (1) is replaced by temperature T , and correspondingly, the source term R should be the viscous heat dissipation and is given by [47,48]

$$R = \frac{2\nu}{C_v} (\mathbf{S}:\mathbf{S}), \quad (58)$$

where $\mathbf{S} = [\nabla \mathbf{u} + (\nabla \mathbf{u})^T]/2$ is the strain rate tensor and C_v is the specific heat at constant volume.

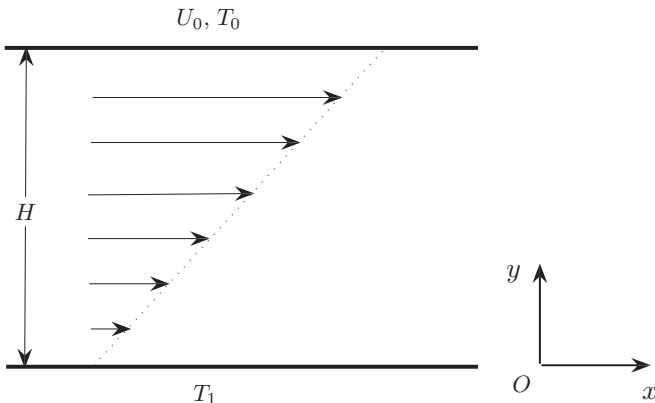


FIG. 4. Schematic of the two-dimensional thermal Couette flow.

The problem has the following analytical solutions of velocity $\mathbf{u} = (U, V)$ and temperature,

$$U(y) = U_0 \frac{y}{H}, \quad V = 0, \quad (59)$$

$$\frac{T - T_1}{T_0 - T_1} = \frac{y}{H} + \frac{\text{Pr Ec}}{2} \frac{y}{H} \left(1 - \frac{y}{H}\right), \quad (60)$$

where $\text{Pr} = \nu/D$ is the Prandtl number and $\text{Ec} = U_0^2/[C_v(T_0 - T_1)]$ is the Eckert number. From Eqs. (59) and (60), we can also obtain the heat flux $\mathbf{J} = (J_x, J_y)$,

$$\begin{aligned} J_x &= -D \frac{\partial T}{\partial x} + T U \\ &= U_0 \frac{y}{H} \left\{ T_1 + (T_0 - T_1) \frac{y}{H} \left[1 + \frac{\text{Pr Ec}}{2} \left(1 - \frac{y}{H}\right) \right] \right\}, \end{aligned} \quad (61)$$

$$\begin{aligned} J_y &= -D \frac{\partial T}{\partial y} + T V \\ &= -D \frac{(T_0 - T_1)}{H} \left[1 + \frac{\text{Pr Ec}}{2} \left(1 - \frac{2y}{H}\right) \right]. \end{aligned} \quad (62)$$

The problem is a little more complicated than Example 1 considered above not only because there is a source term in the CDE, but also because it is coupled with the N-S equations (3) and (4) through velocity. For this reason, in addition to the present MRT model for the CDE, another lattice Boltzmann model to solve the N-S equations is also needed. In this paper, the MRT model, first proposed by d'Humières [30] and then further developed by Lallemand and Luo [31], is adopted. Similar to Eq. (5), the evolution equation of the MRT model for fluid flow reads

$$\begin{aligned} f_i(\mathbf{x} + \mathbf{c}_i \delta t, t + \delta t) - f_i(\mathbf{x}, t) \\ = -(\mathbf{M}^{-1} \mathbf{\Gamma} \mathbf{M})_{ij} [f_j(\mathbf{x}, t) - f_j^{(\text{eq})}(\mathbf{x}, t)] + F_i(\mathbf{x}, t), \end{aligned} \quad (63)$$

where $f_i(\mathbf{x}, t)$ is the distribution function and $f_i^{(eq)}(\mathbf{x}, t)$ is its equilibrium distribution function and is defined as [31]

$$f_i^{(eq)} = w_i \left\{ \rho + \rho_0 \left[\frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right] \right\}, \quad (64)$$

where ρ is the fluid density. $\mathbf{\Gamma}$, similar to $\mathbf{\Lambda}$ in Eq. (12), is also a relaxation matrix and is given as

$$\mathbf{\Gamma} = \text{diag}(\gamma_0, \gamma_1, \dots, \gamma_8). \quad (65)$$

F_i is the discrete forcing term and is given by [7,36]

$$F_i = \left[\mathbf{M}^{-1} \left(\mathbf{I} - \frac{\mathbf{\Gamma}}{2} \right) \mathbf{M} \right]_{ij} \bar{F}_j, \quad (66)$$

where $\bar{F}_i(\mathbf{x}, t) = w_i \delta t \left[\frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})}{c_s^2} \mathbf{c}_i \right] \cdot (\rho_0 \mathbf{a})$ is related to acceleration $\mathbf{a}$ induced by an external force.

The macroscopic density and velocity can be computed by

$$\rho = \sum_i f_i, \quad \mathbf{u} = \frac{1}{\rho_0} \sum_i \mathbf{c}_i f_i + \frac{\delta t}{2} \mathbf{a}. \quad (67)$$

Through the Chapman-Enskog expansion [36], we can obtain the incompressible N-S equations (3) and (4) with the kinematic viscosity $\nu = c_s^2(1/\gamma_{7,8} - 1/2)\delta t$.

In addition, we also would like to point out that the source term $\mathbf{R}$ in Eq. (58) can be computed efficiently in the LBM without adopting finite-difference schemes [36],

$$\mathbf{S} = - \frac{\mathbf{c}_i \mathbf{c}_i (\mathbf{M}^{-1} \mathbf{S} \mathbf{M})_{ij} [f_j - f_j^{(eq)}]}{2c_s^2 \delta t} + \frac{(\gamma_{7,8} - \gamma_2)(\mathbf{u} \cdot \mathbf{a}) \mathbf{I} - \gamma_{7,8}(\mathbf{u} \mathbf{a} + \mathbf{a} \mathbf{u})}{4c_s^2}, \quad (68)$$

which can also be expressed by the macroscopic variables in the moment space,

$$S_{xx} = - \frac{s_e(e - e^{(eq)}) + 3s_v(p_{xx} - p_{xx}^{(eq)})}{12c_s^2 \delta t} - \frac{s_e(\mathbf{u} \cdot \mathbf{a}) + s_v(u_x a_x - a_y u_y)}{4c_s^2}, \quad (69a)$$

$$S_{xy} = S_{yx} - \frac{s_v(p_{xx} - p_{xx}^{(eq)})}{2c_s^2 \delta t} - \frac{s_v(u_x a_y + a_x u_y)}{4c_s^2}, \quad (69b)$$

$$S_{yy} = \frac{3s_v(p_{xx} - p_{xx}^{(eq)}) - s_e(e - e^{(eq)})}{12c_s^2 \delta t} + \frac{s_v(u_x a_x - a_y u_y) - s_e(\mathbf{u} \cdot \mathbf{a})}{4c_s^2}. \quad (69c)$$

In our simulations, the physical parameters were set as $H = 1$, $U = 0.05$, $T_0 = 1$, and $T_1 = 0$; the periodic boundary condition is applied in the horizontal direction. For this steady problem, the following convergent criterion was used to ensure that the numerical results reach the steady state:

$$\frac{\sum_{\mathbf{x}} |T(\mathbf{x}, t) - T(\mathbf{x}, t - 100 \times \delta t)|}{\sum_{\mathbf{x}} |T(\mathbf{x}, t)|} < 1.0 \times 10^{-8}. \quad (70)$$

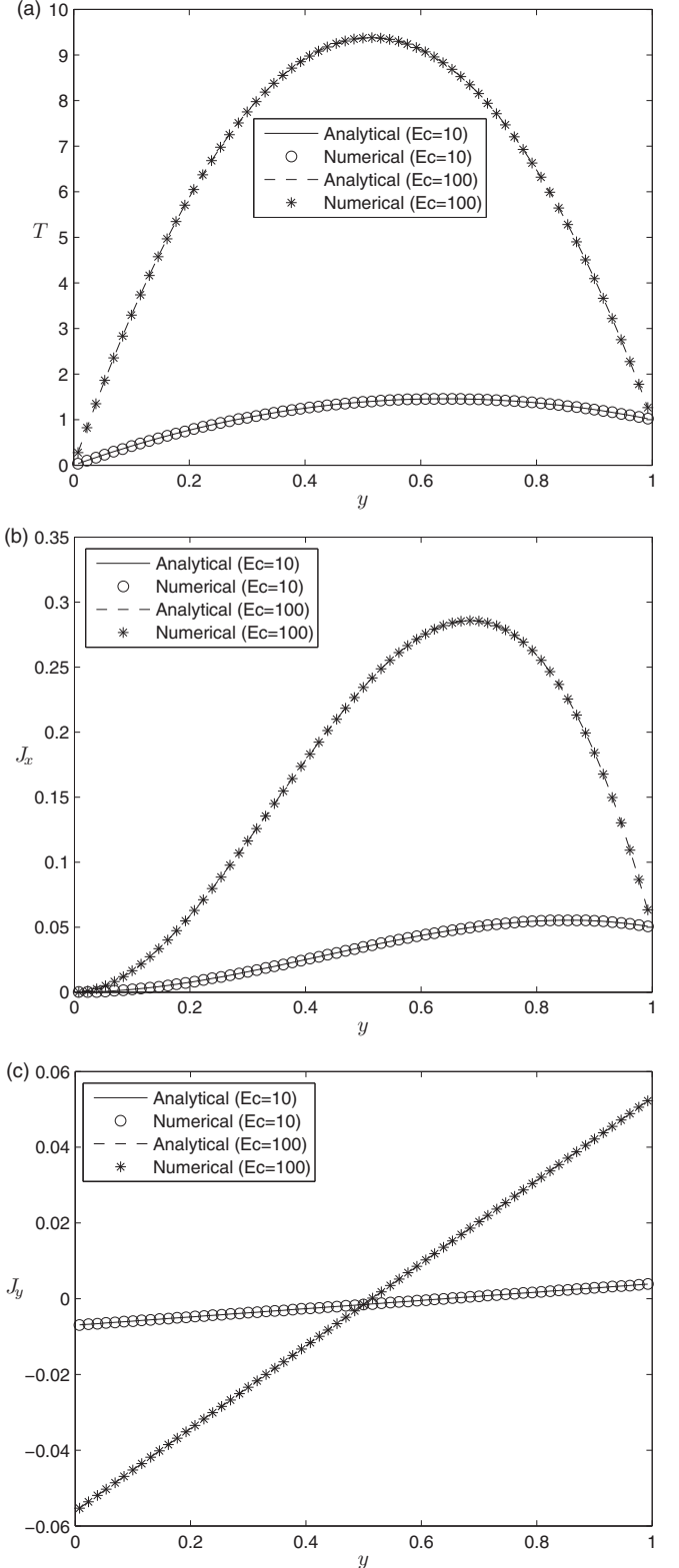


FIG. 5. Comparisons between numerical results and analytical solutions of the thermal Couette flow [(a) temperature T , (b) flux J_x , and (c) flux J_y].

The numerical results are presented in Fig. 5 where the lattice numbers in the vertical direction $N = 64$, $\text{Pr} = 0.71$, and $\text{Ec} = 10$ and 100 are used. The relaxation parameter λ_D related

TABLE I. Relative errors of temperature T and flux $\mathbf{J}$ at different Eckert numbers.

Ec	$E(T)$	$E(J_x)$	$E(J_y)$
10	1.154×10^{-4}	1.241×10^{-4}	1.099×10^{-6}
100	1.964×10^{-4}	2.926×10^{-4}	1.208×10^{-6}

to diffusivity is given as $1/\lambda_D = 0.8$, whereas the one $\gamma_{7.8}$, corresponding to kinematic viscosity ν , is determined from the prescribed Prandtl number. Besides the parameter $\gamma_{7.8}$, the other ones in Eq. (65) are set as $\gamma_0 = \gamma_3 = \gamma_5 = 0.0$, $\gamma_1 = \gamma_2 = 1.0$, and $\lambda_4 = \lambda_6 = 8.0 \times (2.0 - \gamma_{7.8})/(8.0 - \gamma_{7.8})$. As seen from this figure, the numerical results of temperature [Fig. 5(a)] and heat flux [Figs. 5(b) and 5(c)] agree well with the analytical solutions.

To give a quantitative comparison between the numerical results and the analytical solutions, we further computed the relative errors [Eq. (52)] at different Eckert numbers and presented the results in Table I. As shown in this table, the maximum relative error is less than 3.0×10^{-4} , indicating the present MRT model and nonequilibrium scheme are very accurate for the temperature and heat flux. In addition, from Table I one can also observe that relative errors of temperature and the flux increase with an increase in Eckert number. This is because the heat dissipation is more significant at a larger Eckert number, which also results in a stronger nonlinear distribution of temperature. Based on Eq. (2), one can find that the nonlinearity of the temperature also influences the computation of the flux.

3. Example 3. Planar thermal Poiseuille flow

We now consider the planar thermal Poiseuille flow with heat dissipation [50,51]. The schematic of the problem is shown in Fig. 6 where the temperature on the top and bottom walls is kept with T_0 and T_1 , and the fluid in a planar channel is driven by a constant force $\mathbf{F} = \rho(a_x, 0)$.

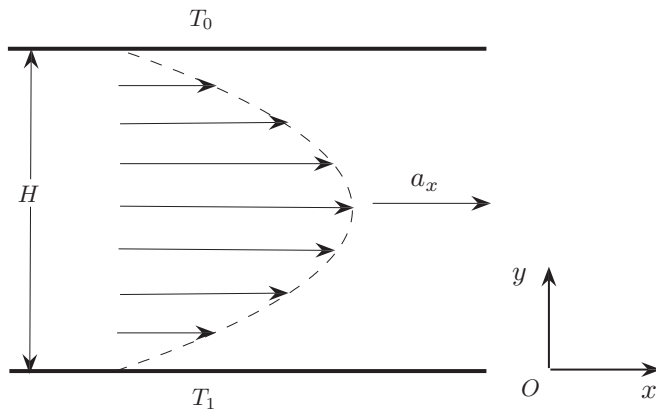


FIG. 6. Schematic of the planar thermal Poiseuille flow.

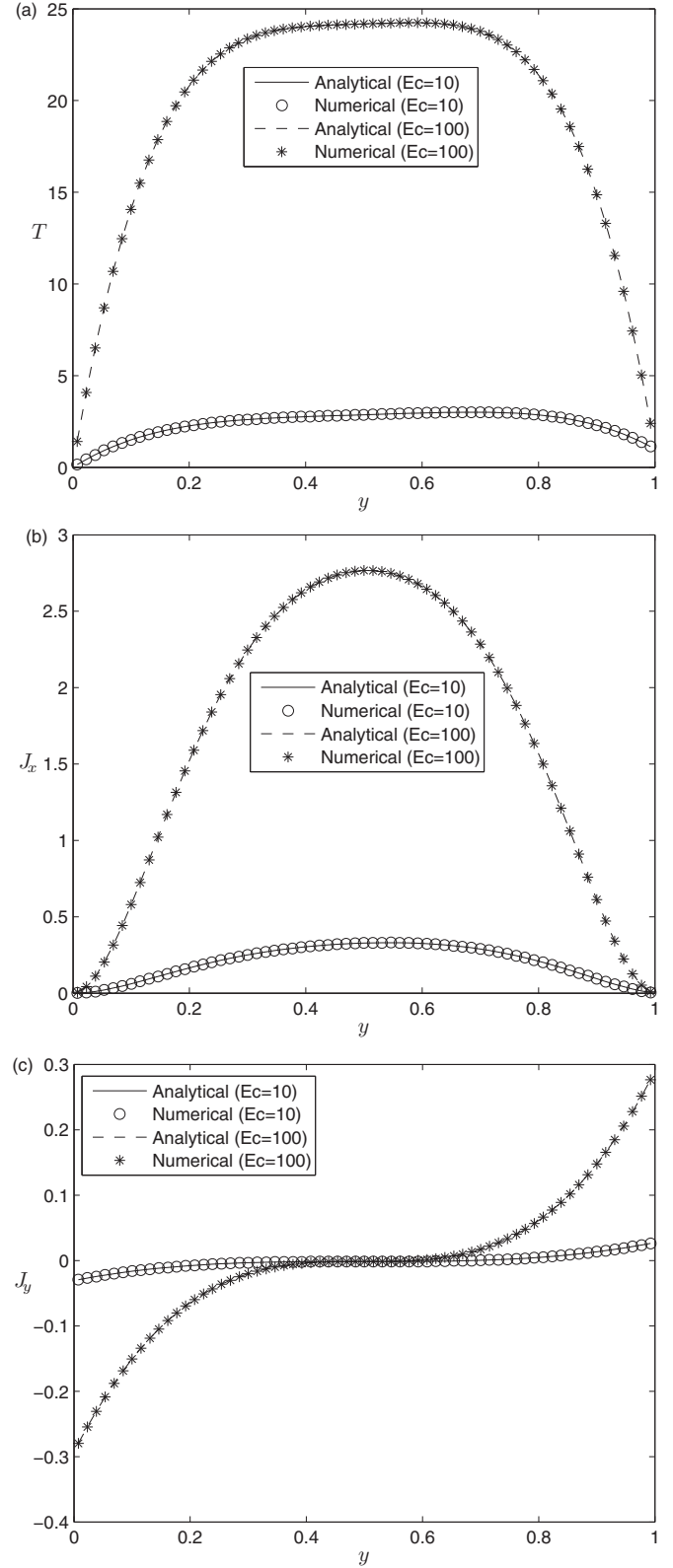
FIG. 7. Comparisons between numerical results and analytical solutions of the thermal Poiseuille flow [(a) temperature T , (b) flux J_x , and (c) flux J_y].

TABLE II. Relative errors of temperature T and flux $\mathbf{J}$ at different Eckert numbers.

Ec	$E(T)$	$E(J_x)$	$E(J_y)$
10	3.398×10^{-4}	2.640×10^{-4}	8.892×10^{-4}
100	4.185×10^{-4}	3.061×10^{-4}	9.395×10^{-4}

The problem is relatively simple and has the exact solutions of velocity, temperature, and heat flux,

$$U(y) = 4U_{\max} \frac{y}{H} \left(1 - \frac{y}{H}\right), \quad V = 0, \quad (71)$$

$$\frac{T - T_1}{T_0 - T_1} = \frac{y}{H} + \frac{\text{Pr Ec}}{3} \left[1 - \left(1 - \frac{2y}{H}\right)^4\right], \quad (72)$$

$$J_x = -D \frac{\partial T}{\partial x} + TU = 4U_{\max} \frac{y}{H} \left(1 - \frac{y}{H}\right) \left(T_1 + (T_0 - T_1) \times \left\{ \frac{y}{H} + \frac{\text{Pr Ec}}{3} \left[1 - \left(1 - \frac{2y}{H}\right)^4\right] \right\}\right), \quad (73)$$

$$J_y = -D \frac{\partial T}{\partial y} + TV = -D \frac{(T_0 - T_1)}{H} \left[1 + \frac{8 \text{Pr Ec}}{3} \left(1 - \frac{2y}{H}\right)^3\right], \quad (74)$$

where $U_{\max} = a_x H^2 / (8\nu)$ is the maximum velocity, $\text{Pr} = \nu/D$ is the Prandtl number, and $\text{Ec} = U_{\max}^2 / [C_v (T_0 - T_1)]$ is the Eckert number. Similar to Example 2, two MRT models are adopted: One is used for the temperature field, and the other is applied for the flow field. The numerical results at different Eckert numbers are presented in Fig. 7 in which the lattice number in the vertical direction, the relaxation parameter, the acceleration due to the external force, and the Prandtl number are set as $N = 64$, $1/\lambda_{7,8} = 0.8$, $a_x = 0.001$, and $\text{Pr} = 0.71$. From this figure one can observe clearly that the numerical results are very close to the analytical solutions.

The relative errors between the numerical results and the analytical solutions are shown in Table II. As seen from the table, the maximum relative error is no more than 1.0×10^{-3} , which indicates that the present MRT model and nonequilibrium scheme are accurate in the study of the temperature and heat flux of the thermal Poiseuille flow. Besides, the larger relative errors of the temperature and the flux are also observed at a larger Eckert number.

B. The convergence rates of the MRT model and nonequilibrium scheme

The theoretical analyses presented in Secs. II and III show that the MRT model for the CDE and the nonequilibrium scheme in computing the flux have a second-order convergence rate in space. This section will numerically validate the theoretical analysis. To this end, the planar thermal Poiseuille flow (Example 3) is still adopted here for the following two reasons. The first is the problem has an exact solution such

that it is more suitable for testing convergence rates of the developed MRT model and nonequilibrium scheme; and the second is that, compared to Examples 1 and 2, Example 3 is more complicated since the forcing term effects should be included in the computations of the strain rate tensor [see Eq. (69)] and the flux [see Eq. (42)]. We carried out some simulations with the same parameters used in Example 3 and computed the relative errors at different lattice sizes. The results in Fig. 8 show that both the present MRT model for the CDE and the nonequilibrium scheme for flux have a second-order convergence rate, which is consistent with our previous analysis.

C. A comparison between the nonequilibrium scheme and the second-order central-difference scheme for the flux

In addition to the nonequilibrium scheme presented above, the finite-difference method can also be used to compute the flux (e.g., Ref. [23]). This section will compare the nonequilibrium scheme and the commonly used second-order central-difference scheme through a study of the planar thermal Poiseuille flow. The second-order central-difference scheme for the flux can be written as

$$J_x(x_i, y_j) = \left(-D \frac{\partial T}{\partial x} + TU\right) \Big|_{(x_i, y_j)} = -D \frac{T(x_{i+1}, y_j) - T(x_{i-1}, y_j)}{2 \delta x} + (TU)_{(x_i, y_j)}, \quad (75)$$

$$J_y(x_i, y_j) = \left(-D \frac{\partial T}{\partial y} + TV\right) \Big|_{(x_i, y_j)} = -D \frac{T(x_i, y_{j+1}) - T(x_i, y_{j-1})}{2 \delta x} + (TV)_{(x_i, y_j)}, \quad (76)$$

where (x_i, y_j) with $1 < j < N$ is the discrete node. Whereas for flux J_y at the fluid nodes nearest to the up and bottom walls, the following second-order finite-difference schemes are adopted

$$J_y(x_i, y_1) = -D \frac{-3T(x_i, y_1) + 4T(x_i, y_2) - T(x_i, y_3)}{2 \delta x} + (TV)_{(x_i, y_1)}, \quad (77)$$

$$J_y(x_i, y_N) = -D \frac{T(x_i, y_{N-2}) - 4T(x_i, y_{N-1}) + 3T(x_i, y_N)}{2 \delta x} + (TV)_{(x_i, y_N)}. \quad (78)$$

Figure 9 shows the relative errors of the nonequilibrium scheme and the central-difference scheme where grid number N ranges from 32 to 128. As seen from this figure, the central-difference scheme, the same as the nonequilibrium scheme, indeed has a second-order convergence rate in computing the flux, as expected; whereas it is interesting to note that the errors of the central-difference scheme are larger than those of the nonequilibrium scheme, especially for flux J_y . Besides that, from the point of view of parallel computing, the local

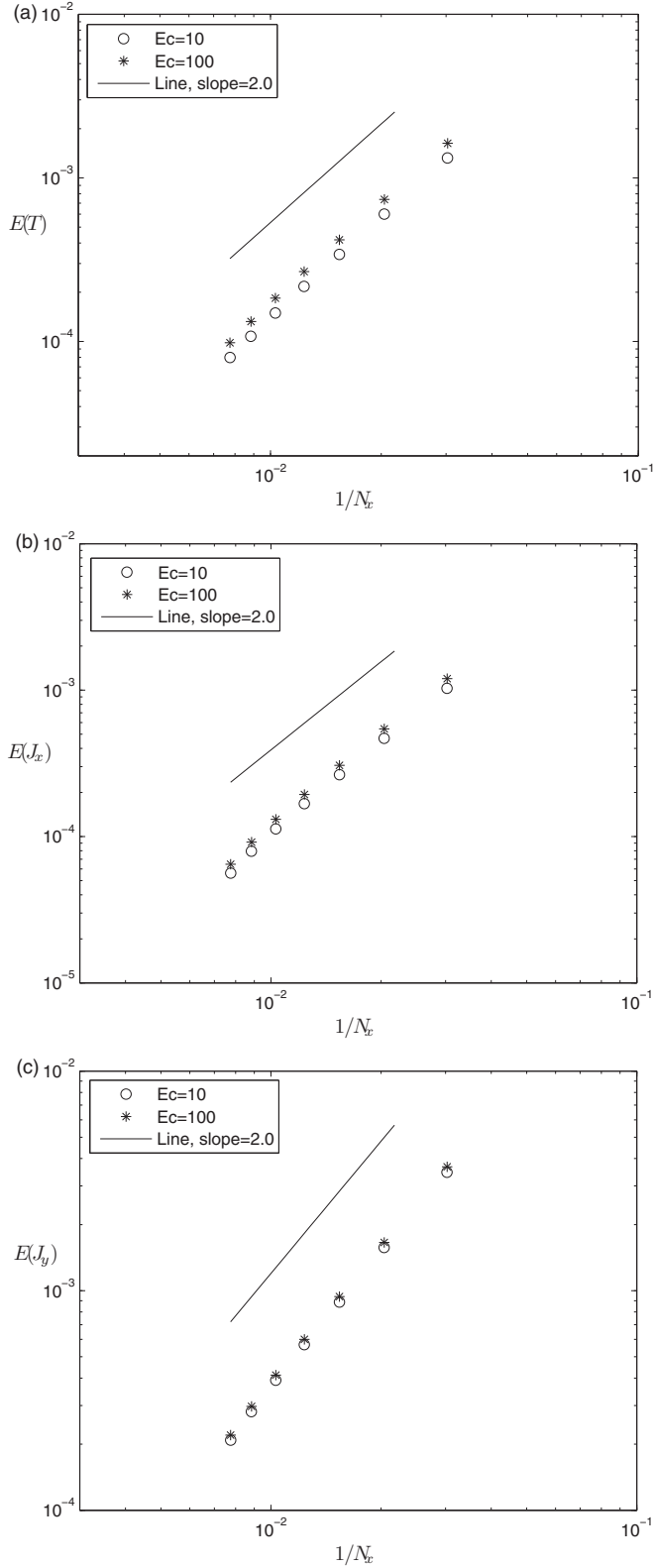


FIG. 8. The relative errors of temperature T and flux $\mathbf{J}$ at different Eckert numbers and lattice sizes ($\delta x = 1/32, 1/48, 1/64, 1/80, 1/96, 1/112$, and $1/128$) [(a) temperature T , (b) flux J_x , and (c) flux J_y]. The slope of the inserted line is 2.0, which is used to indicate that the MRT model and the nonequilibrium scheme have a second-order convergence rate.

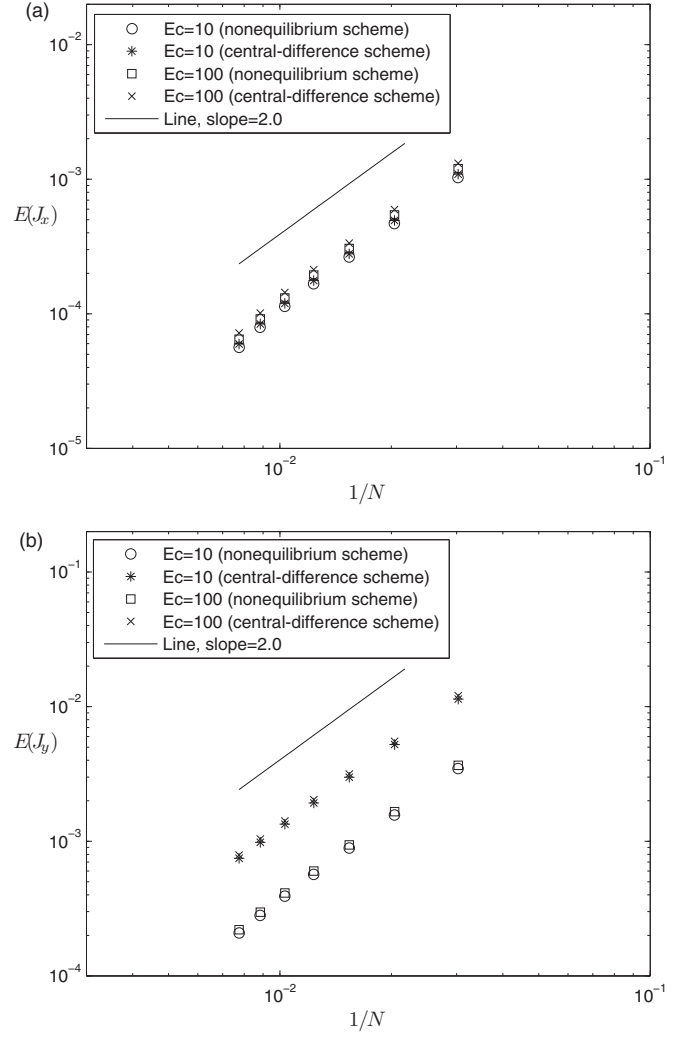


FIG. 9. Comparisons between the nonequilibrium scheme and the second-order central-difference scheme in computing flux [(a) flux J_x and (b) flux J_y]. The slope of the inserted line is 2.0, which is used to indicate that the nonequilibrium scheme and the central-difference scheme have a second-order convergence rate in computing flux.

nonequilibrium scheme is also more efficient than the nonlocal central-difference scheme in computing the flux.

D. A comparison between the BGK and the MRT models

To show the advantages of the MRT model over the BGK model, a comparison between the MRT model and the BGK model was also conducted through using the thermal Couette flow (see Sec. IV B for details). The reason for choosing such a problem is that the thermal Couette flow with viscous heat dissipation is a unidirectional and second-order problem and thus one can readily derive its analytical solutions for the BGK model and the MRT model, which is similar to the procedure reported in a previous paper [46]. To exclude the flow effect, however, here we only consider the CDE for temperature in which the velocity given by Eq. (59) is used

$$\partial_t T + \nabla \cdot (\mathbf{u}T) = D\nabla^2 T + R, \quad (79)$$

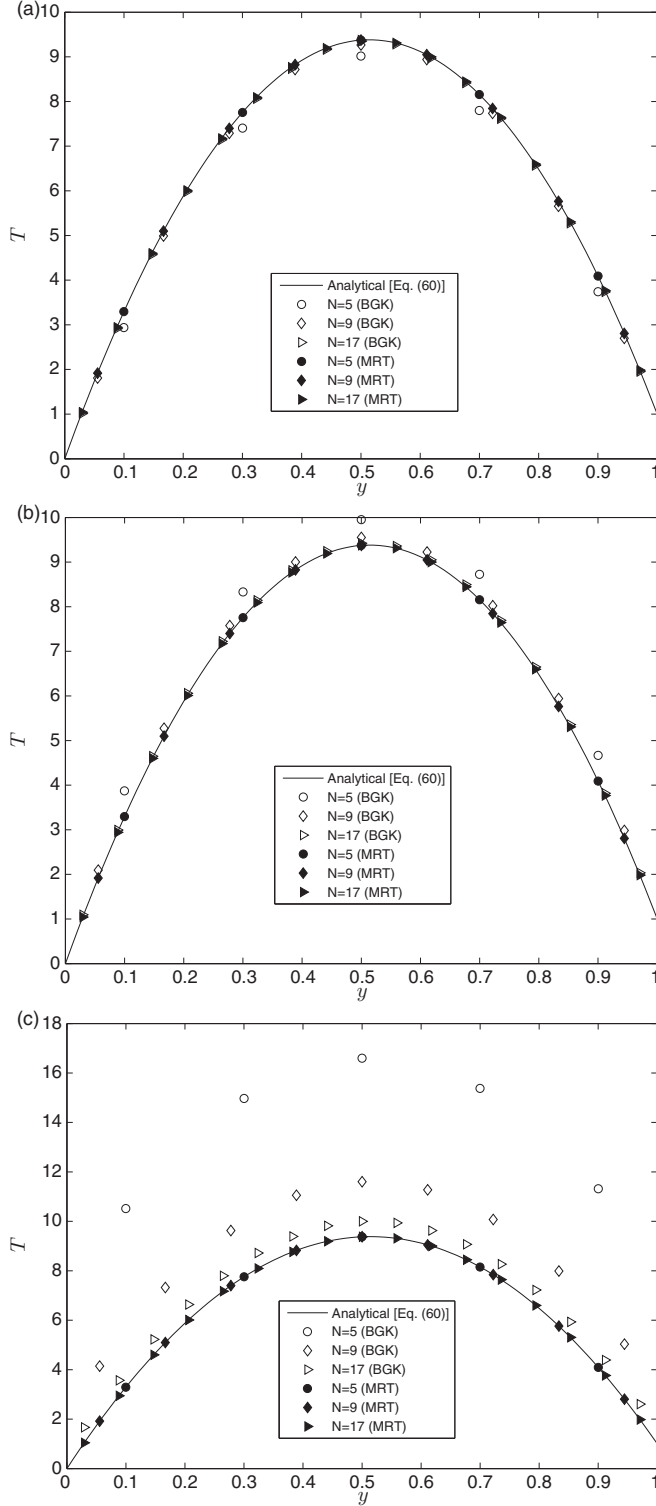


FIG. 10. Comparisons between the BGK and the MRT models for the CDE under different grid numbers and relaxation factors [(a) $1/\lambda_3 = 0.52$, (b) $1/\lambda_3 = 1.2$, and (c) $1/\lambda_3 = 2.5$].

where $R = D \frac{\text{Pr Ec}}{H^2} \Delta T$ with $\Delta T = T_0 - T_1$, the analytical solutions of temperature and flux are still given by Eqs. (60)–(62).

When the BGK model and anti-bounce-back scheme [17,45] are used to solve Eq. (79), one can derive the analytical

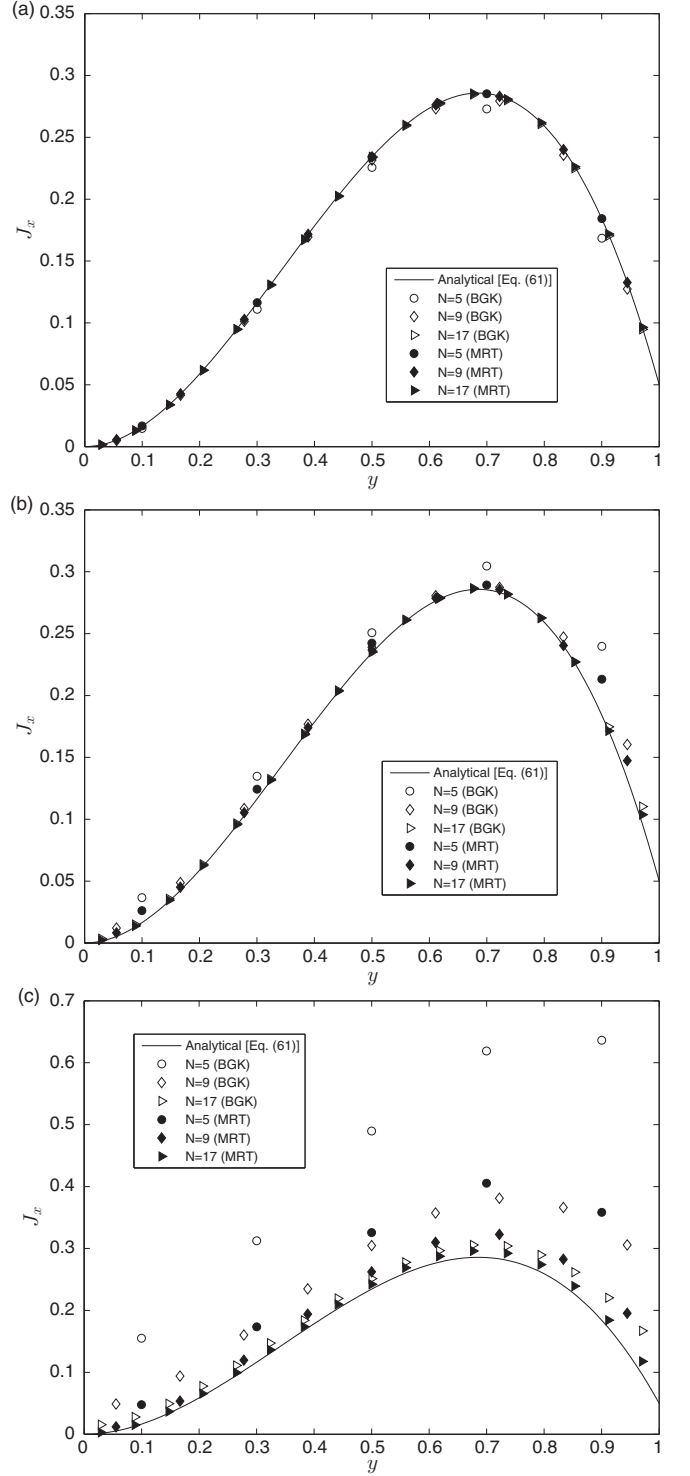


FIG. 11. Comparisons between nonequilibrium schemes of the BGK and MRT models in computing the flux [(a) $1/\lambda_3 = 0.52$, (b) $1/\lambda_3 = 1.2$, and (c) $1/\lambda_3 = 2.5$].

solution of the BGK model,

$$T_j = T_1 + \left[y_j + \frac{\text{Pr Ec}}{2} y_j (1 - y_j) \right] \Delta T + T_s, \quad (80)$$

$$y_j = \frac{j - 0.5}{N},$$

where N is the grid number and T_s is the numerical temperature jump and is given as

$$T_s = \frac{\text{Pr Ec}}{24N^2} \left[4 \left(\frac{2}{\lambda_{\text{BGK}}} - 1 \right)^2 - 3 \right] \Delta T. \quad (81)$$

Compared to the analytical solution of the Couette flow, i.e., Eq. (60), it is clear that a numerical temperature jump T_s , which is proportional to the square of the lattice spacing, is induced in the BGK model and cannot be eliminated unless the relaxation factor λ_{BGK} is set as $\lambda_{\text{BGK}} = 4(2 - \sqrt{3})$. Besides, from Eq. (8), one can also find the BGK model indeed has a second-order convergence rate in space as reported in the previous paper [46].

On the other hand, we can also obtain the analytical solution of the MRT model under the assumption $\lambda_1 = \lambda_7 = \lambda_8$,

$$T_j = T_1 + \left[y_j + \frac{\text{Pr Ec}}{2} y_j (1 - y_j) \right] \Delta T + T_s, \quad (82)$$

$$T_s = \frac{\text{Pr Ec}}{24N^2 \lambda_1 \lambda_3} [\lambda_1 \lambda_3 - 8(\lambda_1 + \lambda_3) + 16].$$

Compared with the results of the BGK model [see Eqs. (80) and (81)], we can find that the temperature jump in the MRT model can be equal to zero if the relaxation factor λ_1 is given by

$$\lambda_1 = \frac{8(\lambda_3 - 2)}{\lambda_3 - 8}. \quad (83)$$

From the above analysis, it is clear that the MRT model would be more accurate than the BGK model through tuning the free relaxation factors. To confirm this statement, we also conducted a comparison between the BGK model and the MRT model in solving Eq. (79) and present the results in Fig. 10 where $\text{Pr} = 0.71$, $\text{Ec} = 100$, the grid number N is varied from 5 to 17, the relaxation factor $1/\lambda_3$ is changed from 0.52 to 2.5, and the other relaxation factors $\lambda_1, \lambda_2, \lambda_4, \lambda_6, \lambda_7$, and λ_8 are determined by Eq. (83).

As seen from Fig. 10, the MRT model can give accurate results, even at a coarse lattice spacing (e.g., $N = 5$), whereas the BGK model cannot, especially at a small λ_3 [see Fig. 10(c)], which is consistent with the above analysis. The results in Fig. 10 also show that, with the increase in the grid number, the difference between the results obtained by the BGK model and the analytical solution becomes small since the numerical temperature jump T_s approaches zero [see Eq. (81)].

In addition, we also presented a comparison between the nonequilibrium schemes of the BGK and MRT models in

computing the flux and show the results in Fig. 11. As shown in this figure, the nonequilibrium scheme of the MRT model is also more accurate than that of the BGK model; this may be because the numerical temperature jump T_s in the BGK model causes an error in computing the flux. Besides, one can also find that the nonequilibrium scheme of the MRT model seems more accurate at a large λ_3 .

V. CONCLUSION

In this paper, we present a local nonequilibrium scheme in the framework of the LBM, as opposed to conventional nonlocal finite-difference schemes, to compute the flux of the convection-diffusion equation. The theoretical analysis and numerical results show that both the nonequilibrium scheme and the MRT model have a second-order convergence rate in space. A comparison between the nonequilibrium scheme and the second-order central-difference scheme in computing the flux indicates that, although both methods have a second-order convergence rate, the nonequilibrium scheme is more accurate. Besides, the nonequilibrium scheme is also more efficient than the conventional finite-difference scheme for its locality in computing the heat and/or mass flux of complex and large-scale problems. And finally, a comparison between the BGK and the MRT models is also performed, and the results show that the MRT model is more accurate than the BGK model, both in solving the CDE and in computing flux. Because of these advantages, the present MRT model, along with the nonequilibrium scheme, can be used to study many complex heat/mass transfer problems [24,25] to predict effective transport properties for fluid flow in porous media [26,27] and to investigate multiphase flows [52,53].

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